

Hirerachy - Reusability

- | | |
|------------------------|-------------------------|
| 1. Association (has-a) | Dependency -> Dependent |
| 2. Inheritance (is-a) | Parent -> Child |

Rules of Method overrrding

1. Name of method should be same
2. Signature should be same
3. The visibility modifier should be same or it should be of wider type
4. The return type of method should be same or it should be its subtype
5. The exception specification list should be same or it should be its subset

```
int *ptr = (int *)malloc(1*sizeof(int))
```

upcasting

- Storing the object of subclass into the super class reference is called as upcasting
- converting the reference of subclass into the super class reference is called as upcasting
- When upcasting is done the super class reference can only access the members of the super class inheritaed in the subclass
- the super class reference cannot access the members defined by the sub class, this is called as object slicing
- to access the members defined by the subclass the super class reference needs to be converted into the sub class reference.

downcasting

- the conversion of reference of super class into the sub class reference is called as downcasting
- the conversion is only possible if upcasting is done in the same type into which downcasting needs to be done
- if upcasting is not done and we try to downcast it, java throws an exception called as classcastexception
- to avoid this exception we can check for the instance present inside the super class reference by using the operator called as (instance of)

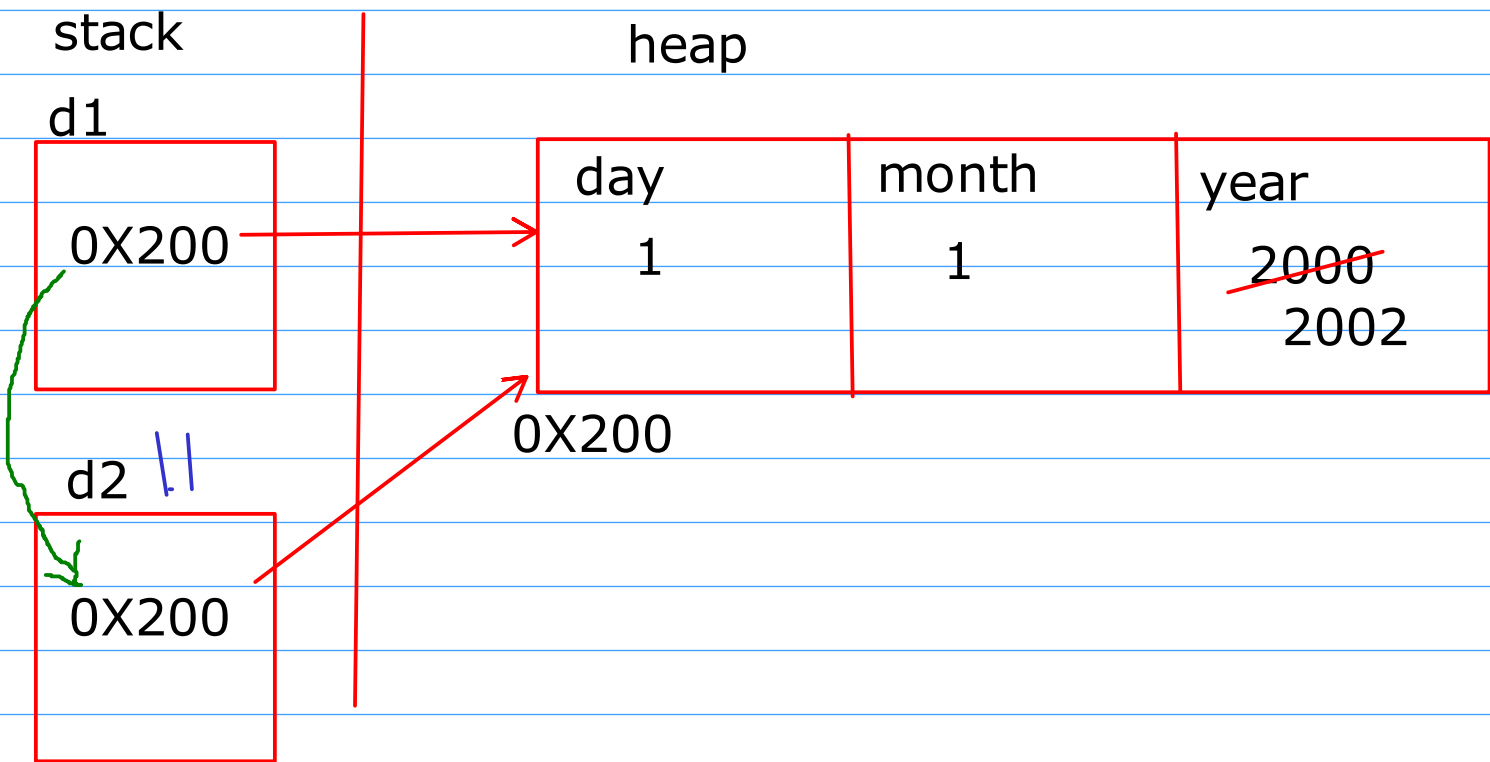
Final keyword

- variable
 - we can declare local variables as final
 - they can be declared and initialized later but once initialized cannot be changed
- fields
 - fields can be declared as final.
 - these must be initialized in field initializer or object initializer or ctor
 - once initialized cannot be changed
- methods
 - If implementation of a method is 100% complete then such methods can be declared as final
 - final methods cannot be overriden
- class
 - If the implementation of a class is 100% complete then such class can be declared as fina
 - Final classes cannot be extended

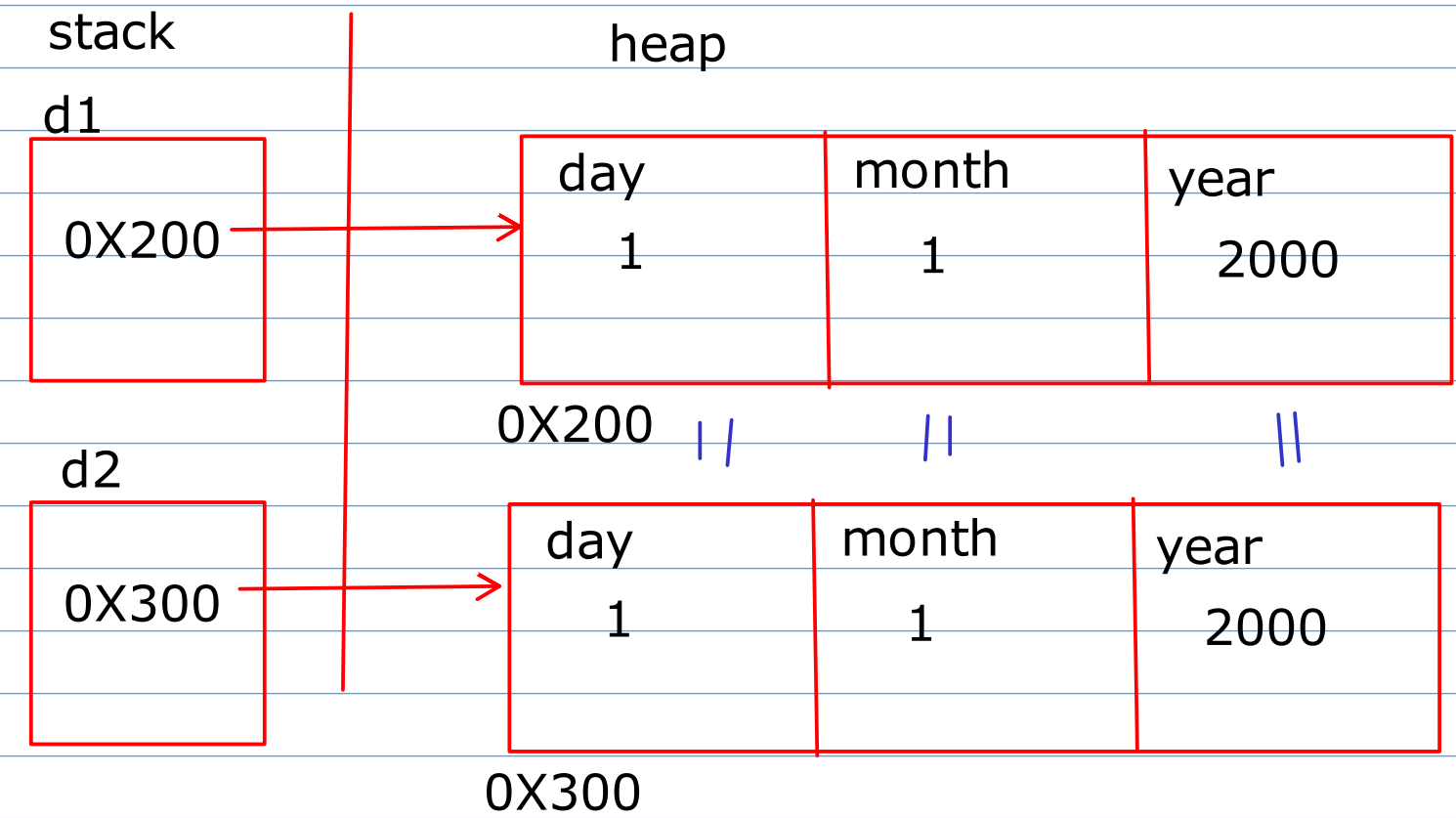
Object

1. toString();

- It is method that prints the state of an object
- It is recommended to override the this method to represent state of an object in text form



```
Date d1 = new Date(1, 1, 2000);
Date d2 = d1;
(d1==d2)-> true
```



```
Date d1 = new Date(1, 1, 2000);
Date d2 = new Date(1 ,1, 2000);
(d1==d2)->
(d1.equals(d2))
```